Lesson 2.2: Vector databases: Pinecone, ChromaDB and Weaviate

What you learn

- **Embeddings.** An embedding turns text into a list of numbers so similar meanings are near each other.
- **Similarity search.** Find the stored items closest to the question's embedding, often by cosine similarity (the code below shows the idea with tiny vectors).
- **Choosing.** ChromaDB is simple to run locally. Pinecone is a managed service. Weaviate is open source and can be self-hosted or managed.

Tools and resources (official sites, not checked live)

- Pinecone
- Chroma
- Weaviate

ASSIGNMENT

Practice for lesson 2.2

Business problem: Run a similarity search.

Steps

1. Run the tested toy search below.
2. Change the query numbers.
3. See which item wins.

Expected output: A top match for each query.

SOLUTION (ANSWER KEY)

PYTHON

```
import math
def cosine(a, b):
    dot = sum(x*y for x, y in zip(a, b)); na = math.sqrt(sum(x*x for x in a)); nb =
math.sqrt(sum(y*y for y in b))
    return dot/(na*nb) if na and nb else 0.0
def top_k(query_vec, items, k=1):
    scored = sorted(((cosine(query_vec, v), name) for name, v in items.items()), reverse=True)
    return [(name, round(s, 3)) for s, name in scored[:k]]
# toy 3-number "embeddings": [about money, about time, about place]
items = {"refund policy": [0.9, 0.3, 0.0], "class timings": [0.1, 0.95, 0.1], "campus
address": [0.0, 0.1, 0.95]}
```

Real embeddings have hundreds or thousands of numbers, but the ranking works the same way.

Check yourself

1. What does an embedding capture?
   Answer: Meaning, as numbers.

2. What does cosine similarity measure?
   Answer: How closely two vectors point the same way.

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